Understanding ADHD Symptoms and Neurobiology in Youth through a Social Determinants of Health Framework

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Background

- Attention-deficit/hyperactivity disorder (ADHD) is a common neurodevelopmental disorder with impact across academic, social, emotional, and family functioning.
  - Estimated prevalence among children in Canada: 5–9% (Espinet et al., 2022)
- Structural magnetic resonance imaging (sMRI) studies have shown numerous neurobiological differences in ADHD, theorizing delayed cortical maturation
- Despite ongoing research, ADHD symptom presentation and neurobiological underpinnings remain heterogeneous
- Social determinants of health (SDoH) factors, including socioeconomic circumstances and family context, may explain differences in ADHD symptoms and neurobiological outcomes.
  - Socioeconomic status: Parental income, education, occupation, and marital status have been associated with ADHD (Russell et al., 2015).
- Pierre Bourdieu's Theory of Social Space provides a framework for organizing SDoH indicators into four forms of capital and constructing quantitative scores for each domain (Bourdieu, 1986).
  - Economic capital: Household income and employment status.
  - Social capital: Household structure and household size.
  - Cultural capital: Caregiver education and language capacity.
  - Symbolic capital: Child ethnic background, racial/ethnic positioning, ethnic identity salience, nativity, and religious affiliation.

Objective

This study applied Bourdieu's Theory of Social Space as a framework to examine how four social determinants of health (SDoH) domains relate to ADHD symptoms and regional cortical thickness in the Adolescent Brain Cognitive Development (ABCD) Study.

Method

Study Sample



- Baseline data from the ABCD study included a total of 1716 participants (9-10 years old); ADHD = 858, non-ADHD = 858
- ADHD diagnosis was classified using the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS) indicator

Figure 1. Pierre Bourdeau's four capitals



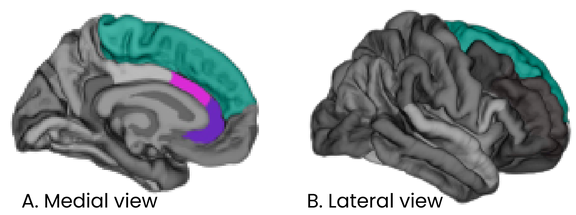
Neuroanatomical

ABCD-provided cortical thickness measures derived using FreeSurfer 7.1.1 and the Desikan–Killiany atlas.
Six regions were examined: bilateral superior frontal gyrus, rostral anterior cingulate cortex, and caudal anterior cingulate cortex

Behavioural

Parent-reported Child Behaviour Checklist (CBCL) subscales: DSM-oriented ADHD, Attention Problems Syndrome Scores, Internalizing and Externalizing T-scores.

Statistical analysis: Pearson correlations examined capital–symptom and capital–cortical thickness associations. Separate multiple linear regressions entered all four capital T-scores simultaneously for each behavioural outcome.



A. Medial view B. Lateral view

Cortical regions of interest. Medial (A) and lateral (B) views illustrating the superior frontal gyrus and rostral and caudal anterior cingulate cortex.

Results

- In multiple regression models including all four capitals (N = 1,714), economic capital remained negatively associated with ADHD symptoms (β = -.168), attention problems (β = -.170), internalizing symptoms (β = -.222), and externalizing symptoms (β = -.242; all $p < .001$). Social, cultural, and symbolic capital were not statistically significant in these models. Together, the four capitals accounted for 2.7%–6.6% of symptom variance.

Figure 1. Associations between SDoH captials and CBCL Symptom Ratings

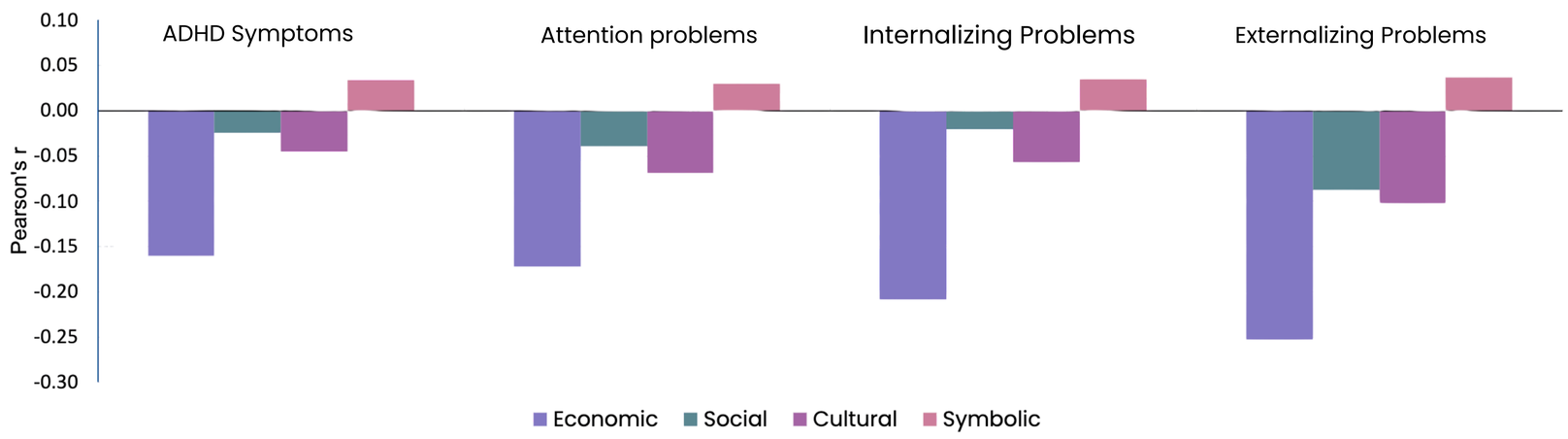
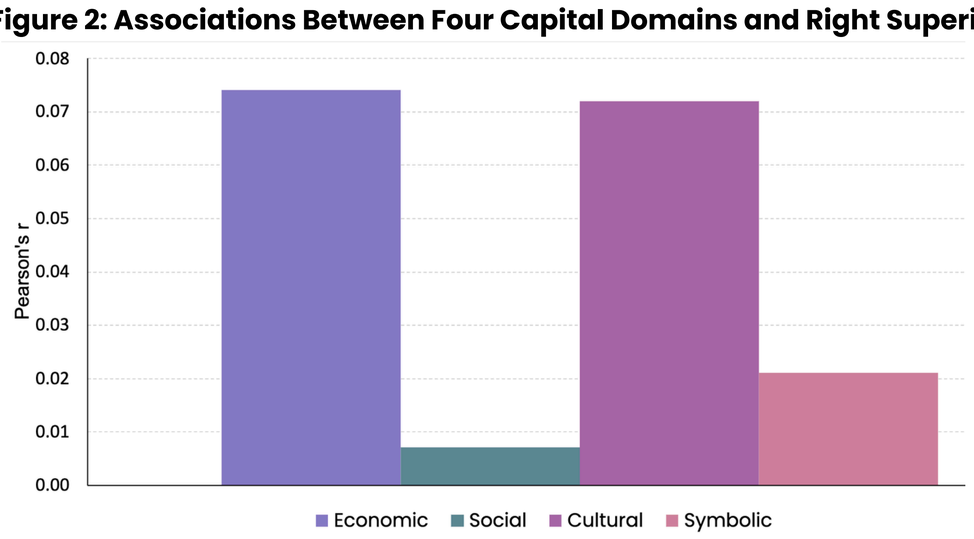


Figure 2: Associations Between Four Capital Domains and Right Superior Frontal Gyrus Cortical Thickness



Bars represent Pearson correlations between economic, social, cultural, and symbolic capital T-scores and right superior frontal gyrus cortical thickness. Positive values indicate that higher capital scores were associated with greater cortical thickness.

- SDoH Capital scores were associated with regional cortical thickness, although correlations were smaller in magnitude than the economic-capital associations with behavioural symptoms.
- The largest positive correlations involved right superior frontal gyrus thickness and economic (r = .074, p = .002) and cultural capital (r = .072, p = .003). These findings are preliminary, with p-values unadjusted for multiple comparisons.

Discussion

Interpretation

- Economic capital was associated with ADHD-related and broader behavioural symptoms, including when the other capital domains were considered simultaneously. The modest variance explained indicates that these scores capture only part of the differences in symptom burden.
- SDoH Capitals and their associations with cortical thickness were smaller in magnitude but should not be interpreted as evidence that social circumstances are unrelated to neuroanatomy. Their robustness requires further investigation with covariate adjustment and correction for multiple comparisons.

Limitations and Future Directions

- These preliminary, cross-sectional associations do not establish causality. Analyses were unadjusted for demographic, site, and family-related factors; reported correlation p-values were not corrected for multiple testing. Capital scores are exploratory, and their interpretation depends on indicator selection and coding.
- Future analyses will assess whether associations persist after covariate adjustment and correction for multiple comparisons, and evaluate their sensitivity to alternative capital scoring approaches, including weighted scores.
- Longitudinal research could examine how social capital domains relate to changes in ADHD symptoms and cortical thickness over time, and whether particular resources buffer the effects of socioeconomic disadvantage.